

A Perturbative Approach to the Ground State of Interacting Fermions in the Mean Field Limit

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Objective

Mathematically rigorous analysis of the ground state energy of interacting fermions in the limit $N \rightarrow \infty$.

Mean Field Limit

The N -body Hamiltonian on $L^2_a(\mathbb{T}^{3N})$ is

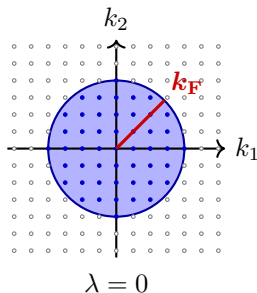
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 $\implies$ **kinetic energy** grows as $\mathcal{O}(N^{5/3})$;

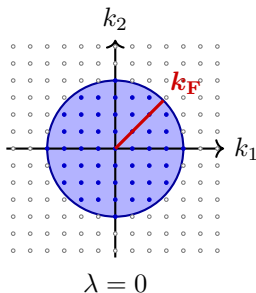


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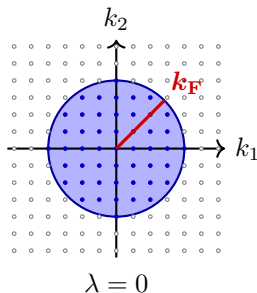


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Mean field scaling

To have an extensive H_N ($H_N \sim N$) in the limit $N \rightarrow \infty$, we define the **effective parameters**

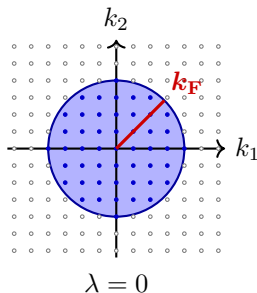
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To have **semiclassical parameter** $\sim N$ in the **mean field coupling constant**

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Main Result

With ordinary second order perturbation theory [[Hainzl-Porta-Rexze, 2019](#)] and then rigorous bosonization methods [[Benedikter-et-al, 2021](#)]:

$$E_N^{\text{gs}} = E_N^{\text{HF}} + E_N^{\text{RPA}} + o\left(N^{-1/3}\right) \quad \text{as } N \rightarrow \infty$$

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As $N \rightarrow \infty$, the leading order contribution to E_N^{gs} is

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Long-term objective: To understand the connection between bosonization and QFT methods.

Sketch of the proof: definitions

Second quantized Hamiltonian:

$$H = \sum_{k \in \mathbb{Z}^3} (\hbar^2 |k|^2 - \mu) a_k^* a_k + \lambda \sum_{p, q, k \in \mathbb{Z}^3} \hat{V}(k) a_{p-k}^* a_{q+k}^* a_q a_p$$

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Duhamel's formula at second order for $\log(Z)$:

$$\begin{aligned} \log \operatorname{Tr} e^{-\beta H} &= \log \operatorname{Tr} e^{-\beta H_0} - \lambda \beta \langle V \rangle_{\beta, H_0} \\ &\quad + \lambda^2 \int_0^\beta \int_0^\tau [\langle V(\tau') V(\tau) \rangle_{\beta, H_0} - \langle V(\tau') \rangle_{\beta, H_0} \langle V(\tau) \rangle_{\beta, H_0}] d\tau' d\tau \end{aligned}$$

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Wick's theorem: $H_0 = \sum_k \epsilon_k a_k^* a_k$ and $A_j = a_j^*, a_j$, then

$$\begin{aligned} \langle A_1 \cdots A_{2n} \rangle_{\beta, H_0} &= \sum_{\sigma \in P_{2n}} (-1)^\sigma \prod_{j=1}^{2n-1} \langle A_{\sigma(j)} A_{\sigma(j+1)} \rangle_{\beta, H_0} \\ \langle A_1 \cdots A_{2n-1} \rangle_{\beta, H_0} &= 0 \end{aligned}$$

where P_{2n} is the set of **pairings** of $2n$ elements.

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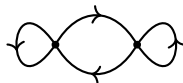
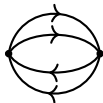
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$$\begin{aligned} \lim_{\beta \rightarrow \infty} \lambda \langle V \rangle_{\beta, H_0} &= \lim_{\beta \rightarrow \infty} \lambda \left(\text{diagram 1} + \text{diagram 2} \right) \\ &= N^{-1} \sum_{\substack{k \in \mathbb{Z}^3, p \in B_F: \\ p-k \in B_F}} \left(\hat{V}(0) - \hat{V}(k) \right) \\ &= \mathcal{O}(N) + \mathcal{O}(1). \end{aligned}$$

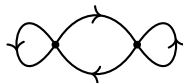
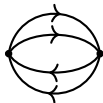
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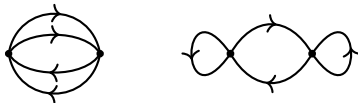
However, they contain unphysical results, like the **tadpole divergence**:

$$\lim_{\beta \rightarrow \infty} \frac{\beta}{2} \left(\text{Diagram} \right)$$

The diagram inside the parentheses is a complex Feynman diagram with two vertices. The left vertex has external lines labeled p and $p-k$. The right vertex has external lines labeled q' and $q'+k'$. Internal lines connect the vertices and form loops, with labels q , $q+k$, p' , $p'-k'$, τ , and τ' indicating specific internal momenta and indices.

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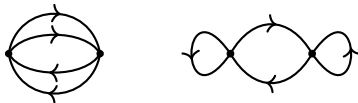
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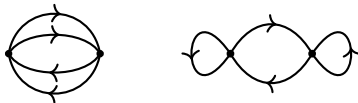
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Remark

Perturbatively, the RPA is understood as a partial resummation **at all orders**. Removing divergences might be highly non-trivial!

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Ultimate goal: to take the proof of the N -scaling of the ground state energy to all orders in perturbation theory.

Thank you!